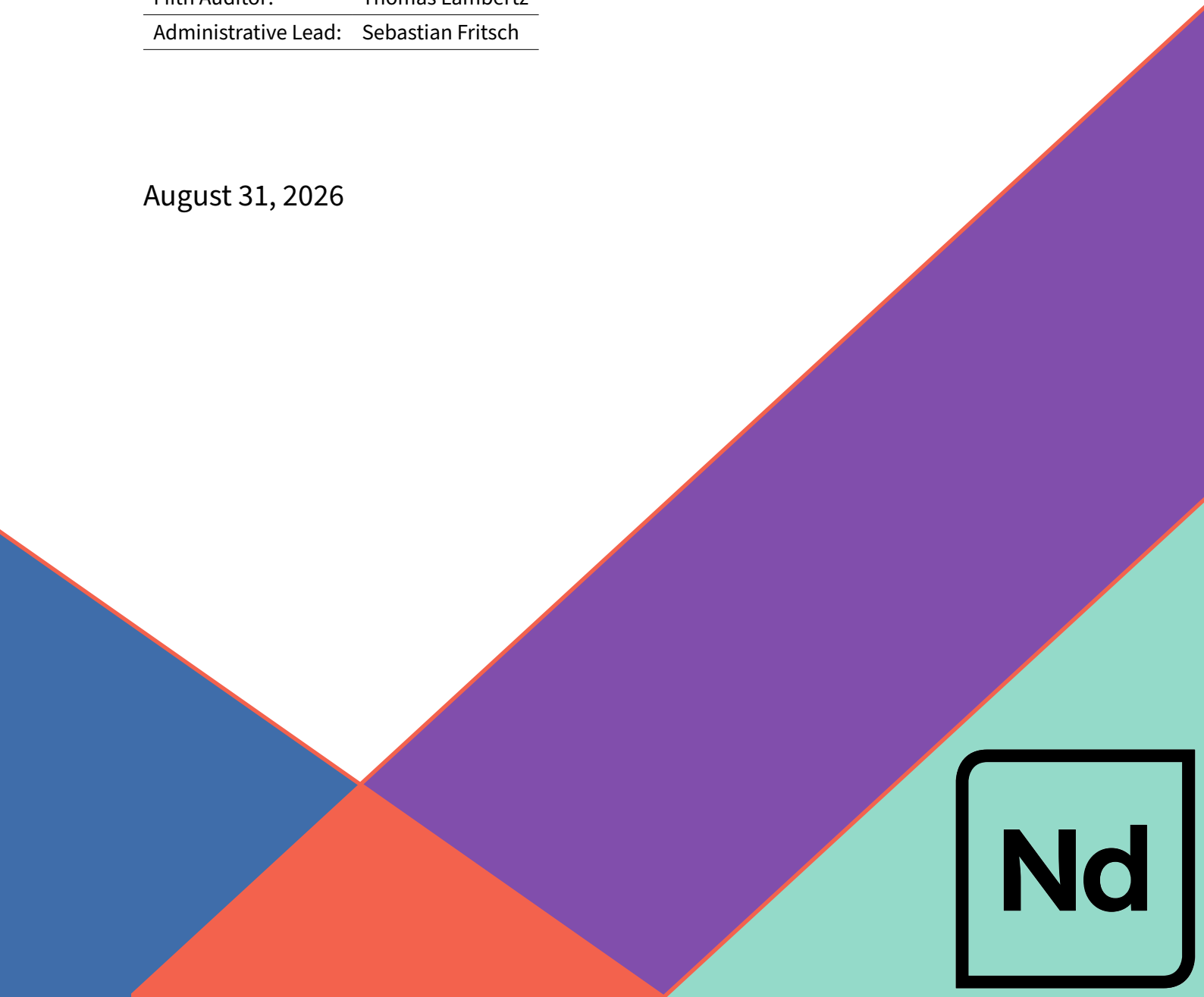

Point Release Audit - Agave v4.2

conducted by Neodyme AG

Lead Auditor:	Sebastian Fritsch
Second Auditor:	Justin Mietzner
Third Auditor:	Tobias Bucher
Fourth Auditor:	Simon Klier
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August 31, 2026



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1 | Executive Summary

On August 17, 2026, Neodyme completed a Point Release Audit of Anza's Agave changes between the v4.1.0 tag (d3f1f55c65) and commit c4b48df969 on the v4.2 branch. The review covered five assigned functional areas. The Alpenglow consensus migration was audited separately and was not part of this audit.

Neodyme identified two nit findings and reported them separately as GitHub issues. Nit findings are not included in this report. Apart from these two nits, no findings were identified. In particular, no critical- or high-severity issues were found.

2 | Scope

Reviewed Revision

Boundary	Reference	Commit
Start	v4.1.0	d3f1f55c65
End	v4.2	c4b48df969

Neodyme reviewed changes to:

- **Transaction Ingestion Pipeline:** receiving transaction packets over QUIC and UDP, deduplicating packets in the signature-verification workers, decoding packets with wincode, limiting memory allocations during decoding, and sending packets through AF_XDP.
- **Leader Schedule & Block Production:** changes to the PoH recorder and tick-hash validation, the replay-stage refactor, low-power mode, and the channels between transaction processing and the PoH recorder.
- **Block Propagation (Turbine/Shreds/XDP):** block-ID repair metadata, XDP error and packet-size handling, shred filtering and chaining, blockstore reads, and the broadcast and retransmit paths.
- **Accounts DB & State Storage:** removal of memory-mapped account storage, separation of the write cache from the accounts index, the accounts lattice hash, root flushing, zero-lamport account handling, and snapshot reconstruction.
- **Gossip & P2P Communication:** the migration of gossip messages from bincode to wincode, decoding limits, signature-verification caches, contact-information parsing, Ping/Pong tokens, and contact-information notifications sent to plugins.

Exclusions

The Alpenglow consensus migration, areas assigned to OtterSec, separately audited cryptography components, and non-production components such as CI, scripts, CLI, RPC, tests, and benchmarks were outside Neodyme's scope. Changes in the areas listed above remained in scope where they interacted with Alpenglow.

3 | Findings

Neodyme identified two nit findings and reported them separately as GitHub issues. Nit findings are not included in this report. Apart from these two nits, no findings were identified. In particular, no critical- or high-severity issues were found.

4 | About Neodyme

Security is difficult.

To understand and break complex things, you need a certain type of people: people who thrive in complexity, who love to play around with code, and who do not stop exploring until they fully understand every aspect of it. That's us.

Our team never outsources audits. Having found over 80 high- or critical-severity bugs in Solana's core code itself, we believe that Neodyme hosts the most qualified auditors for Solana programs. We have also found and disclosed critical vulnerabilities in many of Solana's top projects and responsibly disclosed issues that could have resulted in the theft of over \$10B in TVL on the Solana blockchain.

All of our team members have a background in competitive hacking. During hacking competitions known as CTFs, we competed and collaborated while finding vulnerabilities, breaking encryption, reverse engineering complex algorithms, and much more. Many of our team members have won national and international competitions and continue to rank highly in some of the world's hardest CTF events.

In 2020, some of our members started experimenting with validators and became active members of the early Solana community. After reporting several bugs, the Solana Foundation contracted us for source-code auditing in 2021. As a result, Neodyme was born.

